■SYSTEM CONTROL

1. General

The cruise control has the following control.

Control	Outline	
Constant Speed Control	1GR-FE and 2UZ-FE Engine Models	The ECM compares the actual vehicle speed and the set speed. If the vehicle speed is higher than the set speed, the ECM activates the throttle control motor in the throttle closing direction. If the vehicle speed is lower than the set speed, the ECM activates the throttle control motor in the throttle opening direction.
	1VD-FTV Engine Models	The ECM compares the actual vehicle speed and the set speed. If the vehicle speed is higher than the set speed, the ECM decreases the injection volume by regulating the injectors via the injector drivers. If the vehicle speed is lower than the set speed, the ECM increases the injection volume by regulating the injectors via the injector drivers.
Set Control	While this system fulfils the following conditions, and the cruise control main switch is pressed to the - SET side and released, the ECM stores the vehicle speed and maintains the vehicle constantly at that speed. • ON-OFF button on the cruise control main switch is pressed. • The vehicle is running at a vehicle speed of approximately 40 km/h (25 mph) or more.	
Low Speed Limit Control	The low speed limit is approximately 40 km/h (25 mph). If the vehicle speed drops below that speed while running in the cruise control, the cruise control will be cancelled automatically. However the set speed in the memory is kept.	
COAST (-) Switch Control	While the cruise control main switch is held to the - SET side, the vehicle speed and the set vehicle speed change as follows. • The vehicle decelerates constantly. • The set vehicle speed changes to the speed as which the switch is released.	
Tap Down Control	When the cruise control main switch is pushed momentarily (approximately 0.6 seconds) to the - SET side, the vehicle speed and the vehicle setting speed change as follows. • The vehicle will decelerate in increments of approximately 1.6 km/h (1 mph) for each time the switch is pressed. • However, if the difference between the actual vehicle speed and the vehicle setting speed is greater than 5 km/h (3.1 mph), the vehicle setting speed will change to the speed at which the vehicle was being driven at the time the switch was released.	
ACC (+) Switch Control	When the cruise control main switch is pushed to the + RES side and held, the vehicle speed and the vehicle setting speed change as follows. • The vehicle accelerates constantly. • The set vehicle speed changes to the speed as which the switch is released.	
Tap Up Control	When the cruise control main switch is pushed momentarily (approximately 0.6 seconds) to the + RES side, the vehicle speed and the vehicle setting speed change as follows. • The vehicle accelerates in increments of approximately 1.6 km/h (1 mph) for each time the switch is pressed. • However, if the difference between the actual vehicle speed and the vehicle setting speed is greater than 5 km/h (3.1 mph), the vehicle setting speed does not change.	

(Continued)

Control	Outline
RES Switch Control	If the cruise control is canceled for any reason other than a malfunction (except cancellation by operating the ON-OFF button on the cruise control main switch) and the vehicle speed is more than the low speed limit, the vehicle is returned to the speed set before the cancellation of cruise operation by moving the cruise control main switch to the + RES side. The cruise control mode can be resumed even if the vehicle speed drops below the low speed limit, because the speed in the memory has not been cleared.
Shift Down Control	When the vehicle is cruising uphill, shift down control may be performed by the ECT (Electronic Control Transmission). If the gear has been shifted down, it will be shifted back up after a short time after the ECM judges that the upward slope has finished. If shift down control is performed during ACC (+) or RES switch control, the gear will be shifted back up after the ACC (+) or RES switch control has completed.
Manual Cancel Control	If any of the following signals is sent to the ECM, the cruise control is cancelled accordingly. However, the set speed remains in the memory. • Stop light switch ON signal (Depress the brake pedal) • Shift lever is shifted from D to N *1 • When 1st, 2nd or 3rd range is selected in S mode position *1 • Clutch switch ON signal (Depress the clutch pedal) *2 • CANCEL switch ON signal (cruise control main switch moved to CANCEL side) • Cruise control main switch (ON-OFF button) OFF signal (The set speed in the memory is cleared.)
Automatic Cancel Control	When any of the following conditions occur during cruise control driving, the speed that is set in the memory is cleared and the cruise control is cancelled. • Stop light switch open or short circuit. • Vehicle speed signal malfunction. • Throttle body assembly malfunction (1GR-FE and 2UZ-FE engine models). • Fuel injector malfunction (1VD-FTV engine models). Furthermore, the cruise MAIN indicator light will blink until the ON-OFF button on the cruise control main switch is used to turn the system off, and the operation of the cruise control will be disabled until the ON-OFF button is turned on again.
	When any of the following conditions occur during cruise control driving, the speed that is set in the memory is cleared and the cruise control is cancelled. • Stop light switch input signal is abnormal. • Cancel circuit malfunction. • Cruise control system malfunction (1VD-FTV engine models). Furthermore, the cruise MAIN indicator light will blink until the ON-OFF button on the cruise control main switch is used to turn the system off, and the operation of the cruise control will be disabled until the power source is turned off.
	When the following condition occurs during cruise control driving, the speed that is set in the memory is cleared and the cruise control is cancelled. • Vehicle speed decreases by approximately 16 km/h (10 mph) or more below the speed at which the cruise control was set.
	When the following condition occurs during cruise control driving, the cruise control is cancelled. However, the set speed remains in the memory. • Vehicle speed is below the low speed limit (approximately 40 km/h [25 mph]).
	When the VSC is activated, the cruise control is cancelled. However, the set speed remains in the memory. *3

*1: Only for automatic transmission models
*2: Only for manual transmission models
*3: Only for models with VSC system

2. Diagnosis

If a malfunction occurs in the cruise control system, during cruise control operation, the ECM actuates the automatic cancel control and blinks the cruise MAIN indicator light to inform the driver of a malfunction. At this time, the ECM memorizes the malfunction in the form of 5-digit and 2-digit DTCs (Diagnostic Trouble Code).

- The 5-digit DTC can be read by connecting an intelligent tester II to the DLC3.
- The 2-digit DTC is output to the cruise MAIN indicator light when the TC and CG terminals of the DLC3 connector are connected through the use of the SST (09843-18040). Thus, these DTCs can be obtained by counting the number of blinks of the cruise MAIN indicator light.
- For details, see the Land Cruiser (Station Wagon) Repair Manual (Pub. No. RM0810E).